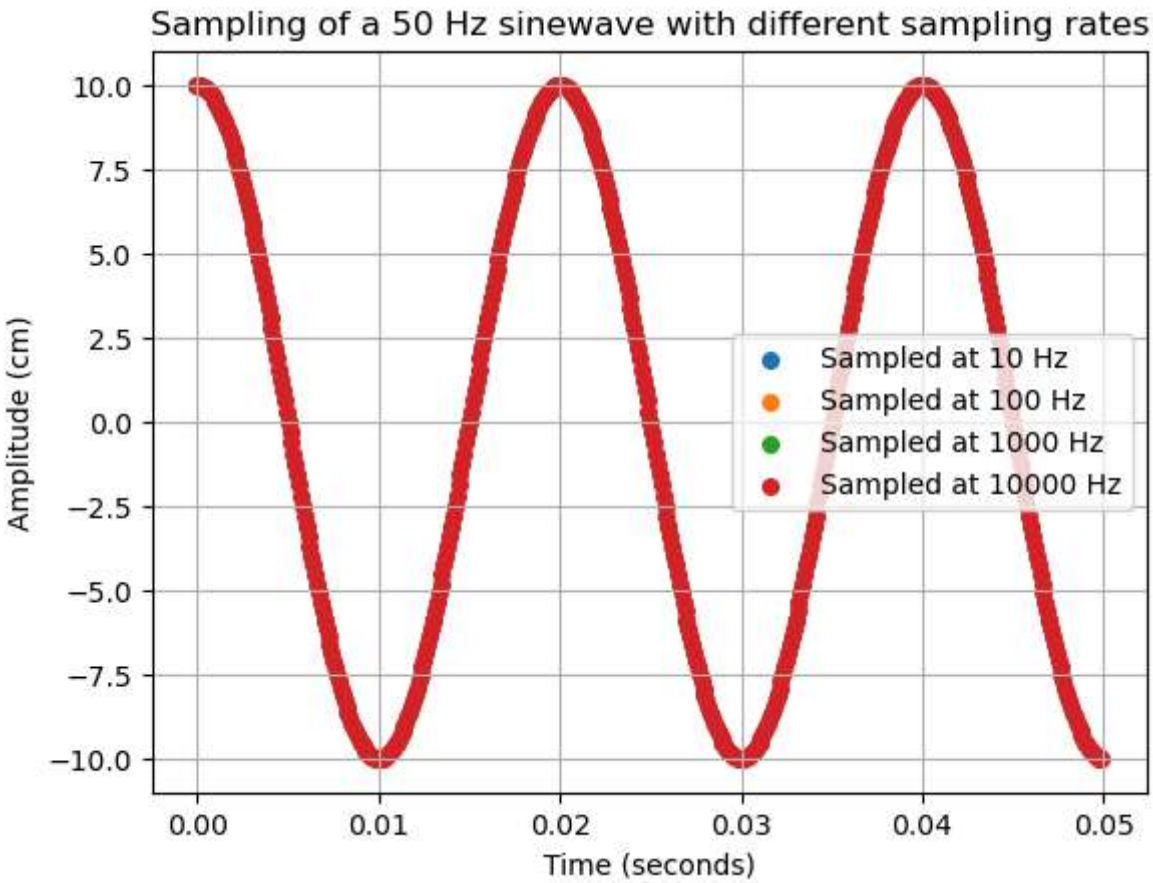
PH3120 – Computational Physics Laboratory 1

CPL106 – Fourier Transform

In [ ]:
```
# Q1 IV).
```

In [7]:
```
import numpy as np
import matplotlib.pyplot as plt

# Given parameters
A = 10  # amplitude in cm
F = 50  # frequency in Hz
pha = np.pi / 2  # phase in radians

# Signal function
def signal(t):
    return A * np.sin(2 * np.pi * F * t + pha)
t_end = 0.05
t_cont = np.linspace(0, t_end, 10000)  # x-axis

# Sampling frequencies
sampling_freqs = [10, 100, 1000, 10000]

# Plot sampled signals
for fs in sampling_freqs:
    t_samples = np.arange(0, t_end, 1/fs)  # X values for scatter
    y_samples = signal(t_samples)    # y values for scatter
    plt.scatter(t_samples, y_samples, label=f'Sampled at {fs} Hz', s=30)

plt.title('Sampling of a 50 Hz sinewave with different sampling rates')
plt.xlabel('Time (seconds)')
plt.ylabel('Amplitude (cm)')
plt.legend()
plt.grid(True)
plt.show()
```


Sampling of a 50 Hz sinewave with different sampling rates
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
Amplitude (cm)
0.00
0.01
0.02
0.03
0.04
0.05
Time (seconds)
Sampled at 10 Hz
Sampled at 100 Hz
Sampled at 1000 Hz
Sampled at 10000 Hz

In [ ]: